

Aditya Kulkarni

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PROFESSIONAL SUMMARY

MS CS candidate (NC State, May 2026) with production experience building **Claude (Anthropic) powered systems** -- multi-turn agentic workflows, MCP tooling, and multi-provider LLM routing across Claude, GPT-4o, and Gemini at Isomer AI. Full-stack across **TypeScript, Node.js, Python, and React**; 394 commits and 7 production deployments. Built a compliance-ready RAG system with per-tenant data isolation for healthcare environments and a vendor-neutral LLM quality harness tracking hallucination rate, latency, and cost/call. Comfortable translating clinical and operational requirements into clean, production-ready AI features.

CORE COMPETENCIES

Claude API & Agentic Workflows

TypeScript & Python

RAG Pipelines

Full-Stack (Node.js / React)

REST APIs & Microservices

AWS & Cloud Infrastructure

Healthcare-Adjacent AI

Production Software Delivery

WORK EXPERIENCE

Isomer AI May 2025 - Present

Product Engineer Intern

- Built production AI features using **Claude (Anthropic), GPT-4o, and Gemini** with MCP tooling -- multi-turn agentic orchestration via **Temporal.io**, multi-provider LLM routing, prompt engineering, and tool use across 8+ models in a live claims automation platform
- Shipped 10+ backend and full-stack features in **TypeScript, Node.js, Express, PostgreSQL, Redis, and React** -- RBAC, real-time event pipelines, document processing (PDF, email, S3), Sentry observability -- **394 commits, 7 production deployments**

Mphasis (HP Company) Jun 2024 - Aug 2024

Software Developer Intern

- Built 5+ AI-driven workflows including an OCR + NLP health claim automation tool -- improved accuracy **25%**; automated deployments on **AWS and GCP**; integrated Grafana monitoring dashboards for production workloads

NC State University Apr 2025 - Present

Graduate Research Assistant

- Designed and deployed a **500GB cloud data platform** on Google Cloud Run for 100+ researchers worldwide; applied hyperparameter tuning and time-series analysis to improve model accuracy 12%

Capgemini Jun 2023 - Aug 2023

AI/ML Intern

- Built distributed **Kafka + Spark** data pipelines reducing ticket-processing latency **20%** and processing time 15%; led end-to-end SDLC for an automated support ticket analysis system

PROJECTS

Glance AI -- Healthcare-Ready RAG System Independent, 2025

Designed a **per-tenant data isolation** architecture at the vector retrieval layer for fintech and healthcare environments -- zero cross-tenant context leakage by design. Deterministic output patterns for factual correctness in regulated data environments; compliance-ready for HIPAA-adjacent use cases. Python · Vector Databases · RAG · LangChain · LLMs

AlgoPulse -- Agentic Orchestration Platform Independent, 2025

Identified API-polling bottleneck under load and re-architected to an event-driven model. **Temporal.io state machines** for durable agentic workflow orchestration; Kafka for fault-tolerant message brokering -- sustained **2,000 async requests/second** with zero message loss under failure injection. TypeScript · Node.js · React · Temporal.io · Kafka · Supabase Edge Functions

LLM Eval Harness -- AI Quality Measurement Independent, 2025

Built a vendor-neutral benchmark comparing **Claude Sonnet 4.6, GPT-4o, and Gemini 2.5 Pro** on a 2,000-example frozen test set. Tracked hallucination rate, abstention accuracy, evidence-quote validity, p50/p95 latency, and cost/call with statistical significance testing. DuckDB + Parquet + Streamlit analytics dashboard. Python · Anthropic SDK · DuckDB · Parquet · Streamlit · Docker

PerCent -- GenAI Finance App HackNC24 Top 10 of 100+ teams, 48h

Shipped a full-stack GenAI application in **48 hours**: goal tracking, expense logging, and a RAG-based AI advisor using Qwen-14B-Chat via Hugging Face -- reduced manual research **30%**. Top 10 of 100+ teams. Python · React · RAG · Hugging Face · REST APIs

EDUCATION

MS Computer Science NC State University

Aug 2024 - May 2026

Artificial Intelligence, Operating Systems, Algorithms, Software Engineering, Data Science, Distributed Systems

B.Tech Computer Science Mumbai University

Aug 2020 - May 2024

CGPA 9.32/10 · Big Data Infrastructure, DS & Algorithms, Machine Learning, Software Engineering

SKILLS

AI & LLMs: Claude API (Anthropic), OpenAI GPT-4o, Gemini, MCP, prompt engineering, agentic workflows, tool use, RAG, LangChain, vector databases
Languages & Frameworks: TypeScript, Python, JavaScript, Node.js, React, Next.js, Express, SQL
Backend & Cloud: REST APIs, microservices, AWS (S3, Lambda, EC2), GCP, Docker, Kubernetes, CI/CD, PostgreSQL, Redis, Prisma
Observability & Infra: Sentry, Grafana, Temporal.io, Kafka, Terraform, GitHub Actions
Awards: Top 10 in 6+ national hackathons · Google Developer Student Club Lead (500+ students) · IEEE Published